

Michael Pham

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EDUCATION

- **University of California, Berkeley**

B.A. in Computer Science and Mathematics

Minor in Data Science

Berkeley, CA

Aug 2022 – Present

- GPA: 3.85
- Member of Upsilon Pi Epsilon Honor Society
- Dean's List, Honors to Date

PROFESSIONAL EXPERIENCE

- **Self-Employed**

Calculus Tutor

West Sacramento, CA

June 2025 – Present

- Tutored Calculus I and II students through personalized lessons.
- Developed practice materials and examples to reinforce core concepts.

PROJECTS

- **Paladins Tracker** | Python, MongoDB

- Built Discord bot to track and report player statistics in real time via external APIs.
- Designed MongoDB schema to store historical and live player data, enabling efficient querying across users.
- Utilized asynchronous requests to handle concurrent user queries at scale.
- Optimized API usage, caching strategies, and database queries, achieving up to 4x speedup across endpoints.

- **Optimizing Convolutions** | C, OpenMP, OpenMPI, SIMD

- Implemented and optimized 2D convolution using parallelism and vectorization, achieving a 15x speedup.
- Improved cache locality through loop tiling and memory access optimization, reducing runtime bottlenecks.
- Applied low-level optimizations including loop unrolling and pointer arithmetic for efficient execution.

- **A Secure File Sharing System** | Golang

- Designed secure file-sharing system supporting multi-user access with encryption and integrity checks.
- Implemented symmetric encryption, MACs, and digital signatures to ensure confidentiality and authenticity.
- Built functionality for sharing, revocation, and integrity verification across distributed users.

- **Real-time Cloth Simulator** | C++

- Built a real-time cloth simulator using mass and spring based system.
- Implemented collision detection and response for self-collisions and external objects.
- Introduced shaders alongside accurate physics interaction with time-varying winds and under high stress.

- **Implementing the GGH Cryptosystem** | Python

- Implemented GGH lattice-based cryptosystem with key generation, encryption, and decryption.
- Optimized algorithm to support higher-dimensional lattices.
- Improved numerical stability by utilizing sympy, reducing encryption errors.

RELEVANT COURSEWORK

- **Computer Science:** Data Structures, Efficient Algorithms and Intractable Problems, Machine Learning, Operating Systems, Computer Graphics, Computer Security
- **Mathematics:** Analysis, Abstract Algebra, Linear Algebra, Discrete Mathematics, Cryptography, Numerical Analysis
- **Data Science:** Principles and Techniques of Data Science, Probability for Data Science, Ecological Data Analysis

TECHNICAL SKILLS

- **Programming Languages:** C, C++, Golang, Java, MATLAB, Python, R, SQL
- **Frameworks/Libraries:** Numpy, Pandas, PyTorch, scikit-learn
- **Tools:** Docker, git, Linux